



Made by

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953212 Database System and Database System Design

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Project Proposal (Initial idea)

Introduction

Searching for motorsport information in detail can be difficult and time-consuming, as there is no universal motorsport encyclopedia available on the internet. We offer users the opportunity to review and research information about the motorsport match tournament, teams, and other related topics.

We are creating an encyclopedia website for individuals interested in motorsports, including tournaments, racers, cars, teams, and familiar staff.

Business Rule

- Motorsport must have one or more Tournaments.
- Team can have more than one Racer.
- Each team can have one or more Cars.
- A Tournament must have more than one Team participating.
- Each Tournament can have only one Motorsport.
- Each racer can be in one or more teams, but in different Tournaments.
- Each racer can race zero or many Matches.
- Each staff can work for zero or many Matches.

Entities and Attributes

- Win count

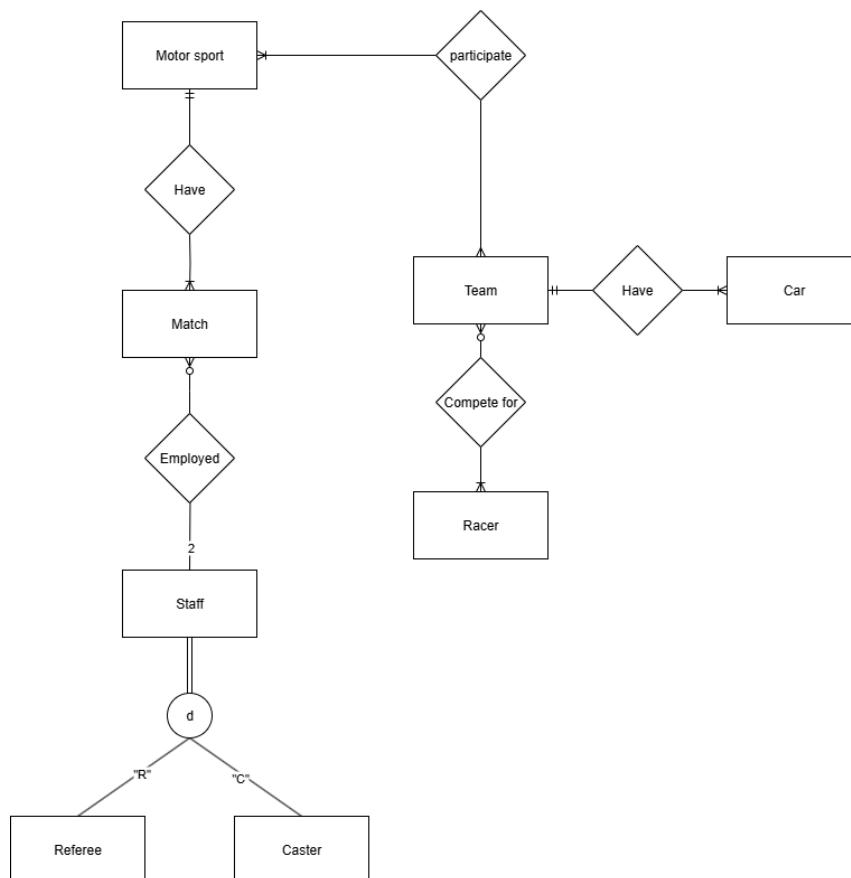
Entities

- Motorsport
 - Motorsport ID
 - Motorsport type
 - Terrain
- Car
 - Car model ID
 - Car type
 - Engine
 - Manufacturer
 - Product Year
- Person
 - Person ID
 - First Name
 - Last Name
 - Status
 - Date of Birth
 - Nationality
- Racer
 - Person ID
 - Racer License
- Team
 - Team's name
 - Team's ID
 - Sponsor
 - Location
- Tournament
 - Viewer Count
 - Tournament ID
 - Team Compete
 - Team ID
 - Avg Laps Time
 - Started time
 - Finished time
 - Placement
 - Date of match
 - Referee
 - Caster
 - Circuit's Address
 - Motorsport ID
- Staff
 - Person ID
 - Years' experience
 - Date of birth
 - Status
 - Nationality
 - Language
 - Referee License

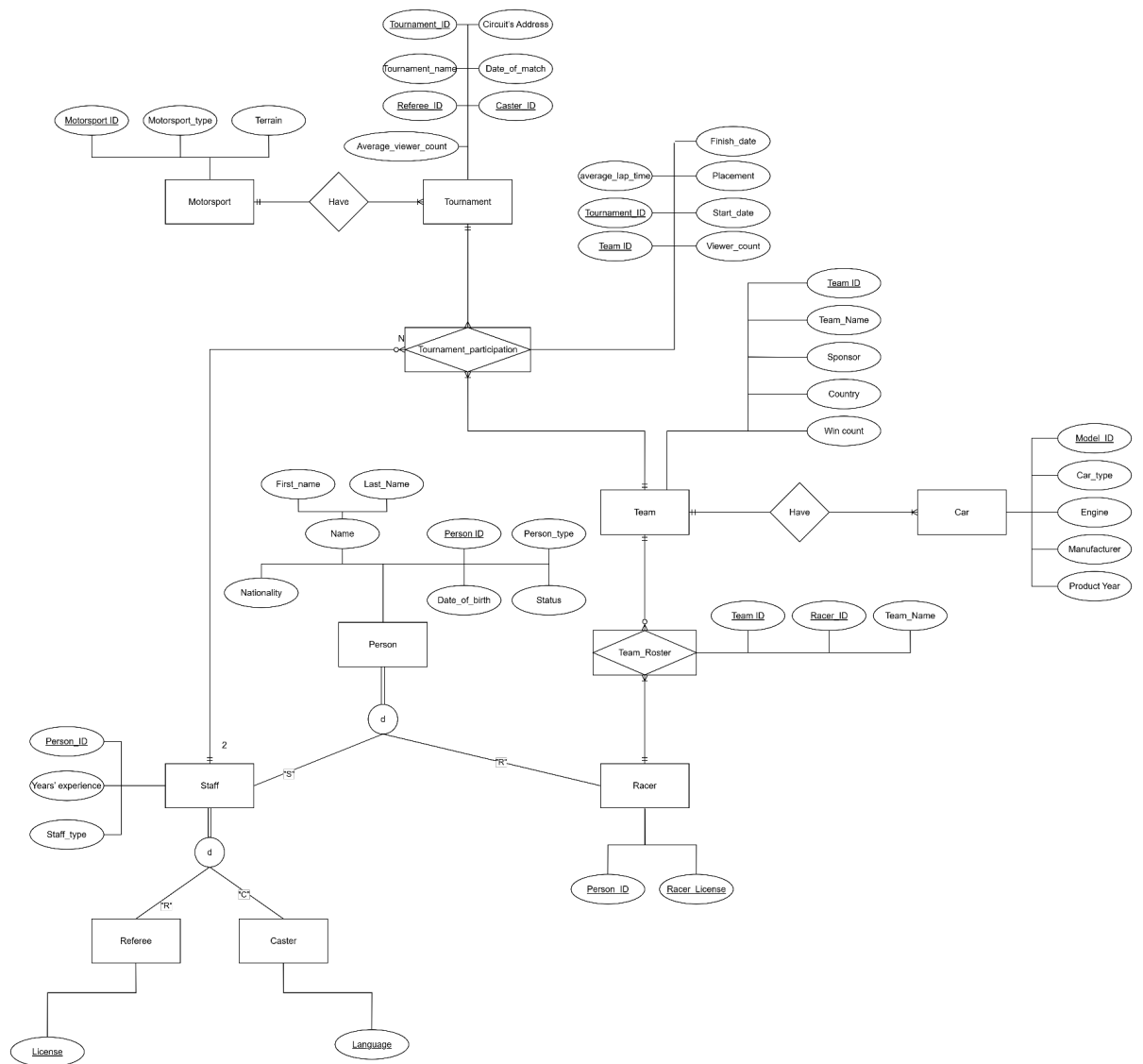
Relationships

- A Motorsport is one-to-many with A Tournament.
- A Tournament is many-to-many with A Team.
- A Car is one-to-many with A Team.
- A Team is many-to-many with A Racer.
- A Staff is many-to-many with A Tournament.

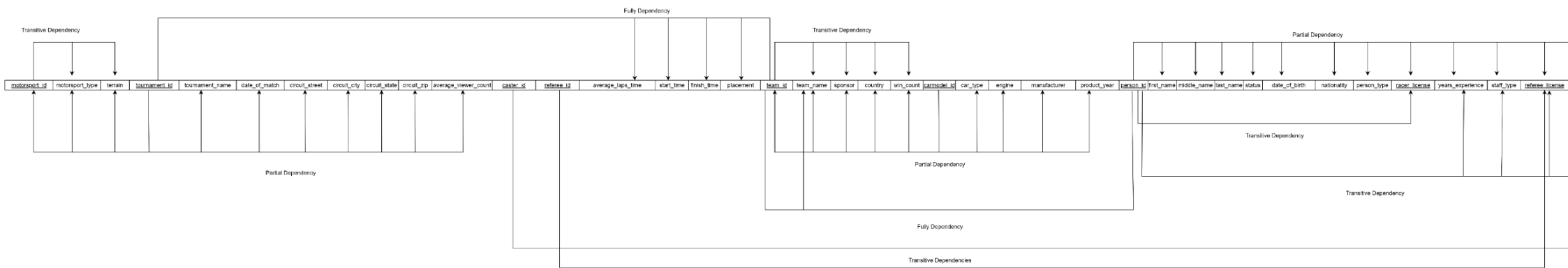
Preliminary ER diagram



Final EER diagram design:



Convert relations into Functional Dependency 1NF



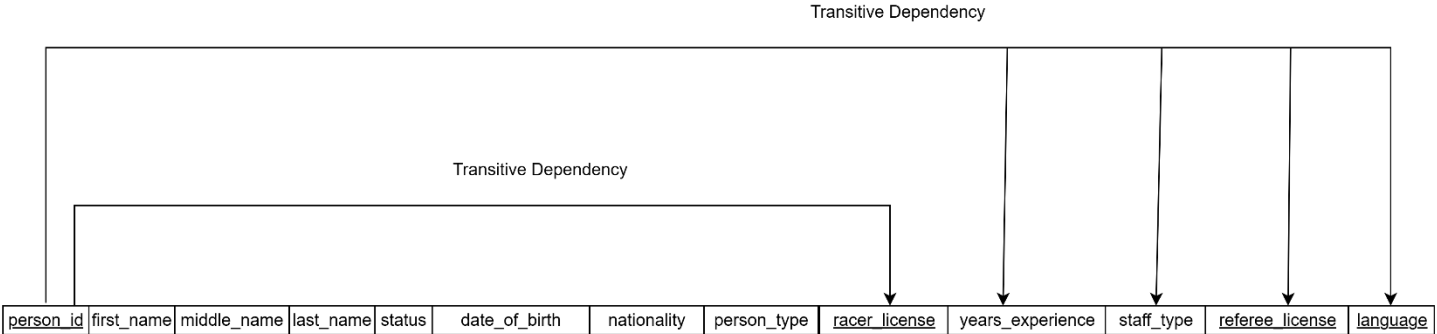
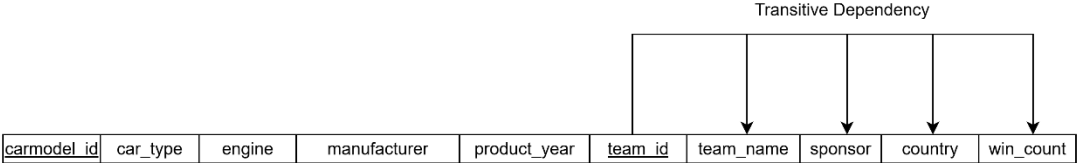
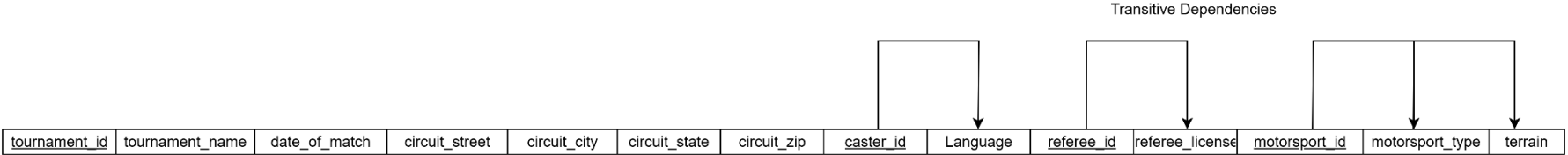
Convert from **1NF** to **2NF**

Partial Dependencies

- Tournament_id → tournament_name, date_of_match, circuit_steet, circuit_city, circuit_state, circuit_zip, caster_id, language, referee_id, referee_license, average_viewer_count, motorsport_id, motorsport_type, terrain
- carmodel_id → team_id, team_name, sponsor, country, win_count, car_type, engine, manufacturer, product_year
- person_id → first_name, middle_name, last_name, status, date_of_birth, nationality, person_type, racer_license, years_experience, staff_type, referee_license, language

Fully Dependencies

- Tournament_id, team_id → average_laps_time, start_time, finish_time, placement
- team_id, person_id → team_name



<u>tournament_id</u>	<u>team_id</u>	<u>person_id</u>	laps_time	start_date	finish_date	placement	viewer_count
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<u>team_id</u>	<u>person_id</u>	team_name
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Convert from 2NF to 3NF

- **Tournament**(Tournament_id, tournament_name, date_of_match, circuit_street, circuit_city, circuit_state, circuit_zip, average_viewer_count, caster_id, referee_id, motorsport_id)
- **Motorsport**(motorsport_id, motorsport_type, terrain)
- **Car**(carmodel_id, car_type, engine, manufacturer, product_year, team_id)
- **Team**(team_id, team_name, sponsor, country, win_count)
- **Person**(person_id, *first_name*, *middle_name*, *last_name*, *status*, *date_of_birth*, *nationality*, *person_type*)
- **Racer**(person_id, racer_license)
- **Staff**(person_id, years_experience, staff_type, referee_license, language)
- **Tournament_participating**(Tournament_id, team_id, average_laps_time, start_time, finish_time, placement)
- **Team_Roster**(team_id, person_id, team_name)

tournament

<u>tournament_id</u>	tournament_name	date_of_match	circuit_street	circuit_city	circuit_state	circuit_zip	average_viewer_count	..caster_id	referee_id	motorsport_id
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Motorsport

<u>motorsport_id</u>	motorsport_type	terrain
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Car

<u>carmodel_id</u>	car_type	engine	manufacturer	product_year	..team_id
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Team

<u>team_id</u>	team_name	sponsor	country	win_count
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Person

<u>person_id</u>	first_name	middle_name	last_name	status	date_of_birth	nationality	person_type
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Racer

<u>person_id</u>	<u>racer_license</u>
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Staff

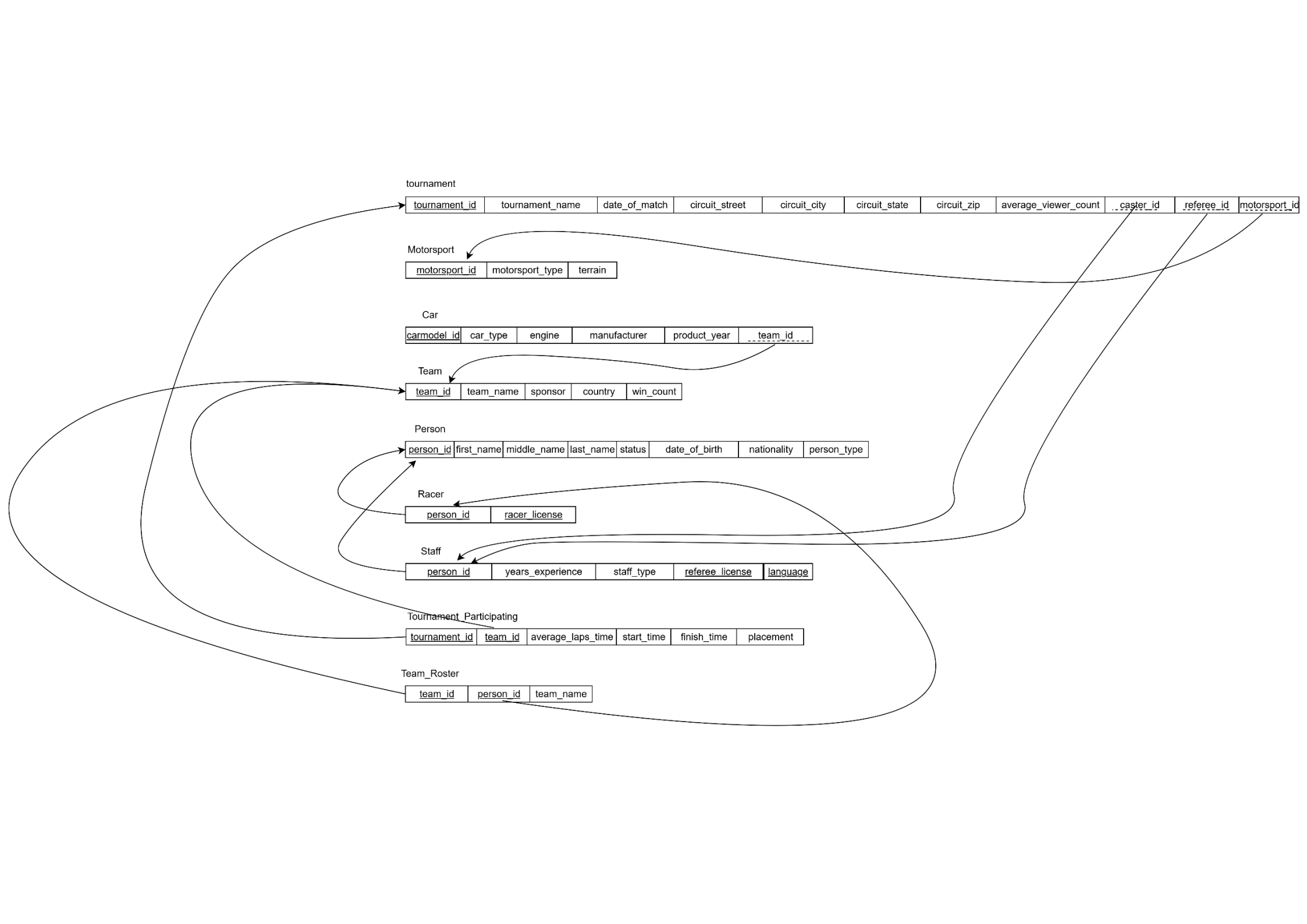
<u>person_id</u>	years_experience	staff_type	<u>referee_license</u>	<u>language</u>
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Tournament Participating

<u>tournament_id</u>	<u>team_id</u>	average_laps_time	start_time	finish_time	placement
----------------------	----------------	-------------------	------------	-------------	-----------

Team_Roster

<u>team_id</u>	<u>person_id</u>	team_name
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Domain Table

Entity / Table	Attributes	Description
Motorsport	Motorsport_ID (PK), Rule, Motorsport_Type, Terrain	Defines the motorsport category and its characteristics.
Tournament	Tournament_ID (PK), Tournament_Name, Date_of_Match, Circuit_Street, Circuit_City, Circuit_State, Circuit_Zip, Average_Viewer_Count, Motorsport_ID (FK) Referee_Id (FK), Caster_Id (FK)	Contains information about tournaments and their schedules/locations.
Team	Team_ID (PK), Team_Name, Sponsor, Location, Win_Count	Represents a racing team participating in tournaments.
Car	Model_ID (PK), Car_Type, Engine, Manufacturer, Product_Year	Represents cars used in tournaments.
Person	Person_ID (PK), First_Name, Middle_Name, Last_Name, Status, Date_of_Birth, Nationality, Person_Type	Generalised entity for all people (racers, staff, referees, casters).
Racer	Person_ID (PK, FK), Racer_License	Specialisation of Person entity — represents a racer with a license.
Staff	Person_ID (PK, FK), Years_Experience, Staff_Type	Specialisation of Person — represents a staff member.
Referee	Person_ID (PK, FK), Referee_License	Subclass of Staff — represents referees with a license.
Caster	Person_ID (PK, FK), Language	Subclass of Staff — represents casters with supported language.
Team_Roster	Team_ID (FK), Racer_ID (FK), Team_Name	Junction table linking teams and racers.
Tournament_Participation	Tournament_ID (FK), Team_ID (FK), Average_Laps_Time, Start_Time, Finish_Time	Tracks which teams, racers, staff, and cars participate in, and which tournament.

Final Schema

1. Motorsport

Attribute	Data Type	Key
motorsport_id	INT	PK
motorsport_type	VARCHAR(25)	
terrain	VARCHAR(50)	

2. Team

Attribute	Data Type	Key
team_id	INT(10)	PK
team_name	VARCHAR(50)	
sponsor	VARCHAR(50)	
country	VARCHAR(30)	
win_count	INT(10)	

3. Person

Attribute	Data Type	Key
person_id	INT(10)	PK
first_name	VARCHAR(20)	
last_name	VARCHAR(20)	
status	VARCHAR(15)	
date_of_birth	DATE	
nationality	VARCHAR(30)	
person_type	VARCHAR(5)	

4. Car

Attribute	Data Type	Key
carmodel_id	INT	PK
car_type	VARCHAR(30)	
engine	VARCHAR(50)	
manufacturer	VARCHAR(50)	
product_year	YEAR(4)	
team_id	INT	FK

5. Tournaments

Attribute	Data Type	Key
tournament_id	INT	PK
tournament_name	VARCHAR(50)	
date_of_match	DATE	
circuit_street	VARCHAR(50)	
circuit_city	VARCHAR(50)	
circuit_state	VARCHAR(50)	
circuit_zip	INT(15)	
average_viewer_count	INT	
motorsport_id	INT	FK
caster_id	INT	FK
referee_id	INT	FK

6. Tournament Participating

Attribute	Data Type	Key
tournament_id	INT	FK
team_id	INT	FK
average_laps_time	DECIMAL	
start_time	DATETIME	
finish_time	DATETIME	
placement	INT	

7. Team Roster

Attribute	Data Type	Key
team_id	INT	PK, FK
person_id	INT	PK, FK
team_name	VARCHAR(50)	

8. Staff

Attribute	Data Type	Key
person_id	INT	PK, FK
years_experience	INT(2)	
staff_type	VARCHAR(7)	
referee_license	INT	
language	VARCHAR(15)	

9. Racer

Attribute	Data Type	Key
person_id	INT	PK, FK
racer_license	INT	UNIQUE

MySQL Final Schema:

```
1 DROP DATABASE IF EXISTS motoropedia;
2 CREATE DATABASE IF NOT EXISTS motoropedia;
3 USE motoropedia;
4
5
6 CREATE TABLE motorsport (
7     motorsport_id INT UNIQUE NOT NULL,
8     motorsport_type VARCHAR(25),
9     terrain VARCHAR(50),
10    PRIMARY KEY (motorsport_id)
11 );
12
13 CREATE TABLE team (
14     team_id INT(10) UNIQUE NOT NULL,
15     team_name VARCHAR(50),
16     sponsor VARCHAR(50),
17     country VARCHAR(30),
18     win_count INT(10),
19     PRIMARY KEY (team_id)
20 );
21
22 CREATE TABLE person (
23     person_id INT(10) UNIQUE NOT NULL,
24     first_name VARCHAR(20),
25     last_name VARCHAR(20),
26     status VARCHAR(15),
27     date_of_birth DATE,
28     nationality VARCHAR(30),
29     person_type VARCHAR(5),
30     PRIMARY KEY (person_id)
31 );
32 CREATE TABLE tournaments (
33     tournament_id INT UNIQUE NOT NULL,
34     tournament_name VARCHAR(50),
35     date_of_match DATE,
36     circuit_street VARCHAR(50),
37     circuit_city VARCHAR(50),
38     circuit_state VARCHAR(50),
39     circuit_zip INT(15),
40     average_viewer_count INT,
41     motorsport_id INT,
42     caster_id INT NOT NULL,
43     referee_id INT NOT NULL,
44     PRIMARY KEY (tournament_id),
45     FOREIGN KEY (motorsport_id) REFERENCES motorsport(motorsport_id),
46     FOREIGN KEY (caster_id) REFERENCES person(person_id),
47     FOREIGN KEY (referee_id) REFERENCES person(person_id)
48 );
49
50 CREATE TABLE car (
51     carmodel_id INT UNIQUE NOT NULL,
52     car_type VARCHAR(30),
53     engine VARCHAR(50),
54     manufacturer VARCHAR(50),
55     product_year YEAR(4),
56     team_id INT,
57     PRIMARY KEY (carmodel_id),
58     FOREIGN KEY (team_id) REFERENCES team(team_id)
59 );
60
61 CREATE TABLE racer (
62     person_id INT UNIQUE NOT NULL,
63     racer_license INT UNIQUE NOT NULL,
64     PRIMARY KEY (person_id),
65     FOREIGN KEY (person_id) REFERENCES person(person_id)
66 );
67
68 CREATE TABLE staff (
69     person_id INT UNIQUE NOT NULL,
70     years_experience INT(2),
71     staff_type VARCHAR(7),
72     referee_license INT,
73     language VARCHAR(15),
74     PRIMARY KEY (person_id),
75     FOREIGN KEY (person_id) REFERENCES person(person_id)
76 );
77
78 CREATE TABLE tournament_participating (
79     tournament_id INT NOT NULL,
80     team_id INT NOT NULL,
81     average_laps_time DECIMAL,
82     start_time DATETIME,
83     finish_time DATETIME,
84     placement INT
85 );
86
87 CREATE TABLE team_roster (
88     team_id INT NOT NULL,
89     person_id INT NOT NULL,
90     team_name VARCHAR(50),
91     PRIMARY KEY (team_id, person_id),
92     FOREIGN KEY (team_id) REFERENCES team(team_id),
93     FOREIGN KEY (person_id) REFERENCES person(person_id)
94 )
```

```
DROP DATABASE IF EXISTS motoropedia;
CREATE DATABASE IF NOT EXISTS motoropedia;
USE motoropedia;
```

```
CREATE TABLE motorsport (
    motorsport_id INT NOT NULL AUTO_INCREMENT,
    motorsport_type VARCHAR(25),
    terrain VARCHAR(50),
    PRIMARY KEY (motorsport_id)
);
```

```
CREATE TABLE team (
    team_id INT NOT NULL AUTO_INCREMENT,
    team_name VARCHAR(50),
    sponsor VARCHAR(50),
    country VARCHAR(30),
    win_count INT,
    PRIMARY KEY (team_id)
);
```

```
CREATE TABLE person (
    person_id INT NOT NULL AUTO_INCREMENT,
    first_name VARCHAR(20),
    last_name VARCHAR(20),
    status VARCHAR(15),
    date_of_birth DATE,
    nationality VARCHAR(30),
    person_type VARCHAR(5),
    PRIMARY KEY (person_id)
);
```

```
CREATE TABLE tournaments (
    tournament_id INT NOT NULL AUTO_INCREMENT,
    tournament_name VARCHAR(50),
    date_of_match DATE,
    circuit_street VARCHAR(50),
    circuit_city VARCHAR(50),
    circuit_state VARCHAR(50),
    circuit_zip INT(15),
    average_viewer_count INT,
    motorsport_id INT,
    caster_id INT,
    referee_id INT,
    PRIMARY KEY (tournament_id),
    FOREIGN KEY (motorsport_id) REFERENCES motorsport(motorsport_id) ON DELETE RESTRICT,
    FOREIGN KEY (caster_id) REFERENCES person(person_id) ON DELETE RESTRICT,
    FOREIGN KEY (referee_id) REFERENCES person(person_id) ON DELETE RESTRICT
);
```

```
CREATE TABLE car (
    carmodel_id INT NOT NULL AUTO_INCREMENT,
    car_type VARCHAR(30),
    engine VARCHAR(50),
    manufacturer VARCHAR(50),
```

```
product_year YEAR,  
team_id INT,  
PRIMARY KEY (carmodel_id),  
FOREIGN KEY (team_id) REFERENCES team(team_id) ON DELETE SET NULL  
);
```

```
CREATE TABLE racer (  
    person_id INT NOT NULL,  
    racer_license VARCHAR(20) UNIQUE NOT NULL,  
    PRIMARY KEY (person_id),  
    FOREIGN KEY (person_id) REFERENCES person(person_id) ON DELETE CASCADE  
);
```

```
CREATE TABLE staff (  
    person_id INT NOT NULL,  
    years_experience INT,  
    staff_type VARCHAR(20),  
    referee_license VARCHAR(20),  
    language VARCHAR(15),  
    PRIMARY KEY (person_id),  
    FOREIGN KEY (person_id) REFERENCES person(person_id) ON DELETE CASCADE  
);
```

```
CREATE TABLE tournament_participating (  
    tournament_id INT NOT NULL,  
    team_id INT NOT NULL,  
    average_laps_time DECIMAL(10, 3),  
    start_time DATETIME,  
    finish_time DATETIME,  
    placement INT,  
    PRIMARY KEY (tournament_id, team_id),  
    FOREIGN KEY (tournament_id) REFERENCES tournaments(tournament_id) ON DELETE CASCADE,  
    FOREIGN KEY (team_id) REFERENCES team(team_id) ON DELETE CASCADE  
);
```

```
CREATE TABLE team_roster (  
    team_id INT NOT NULL,  
    person_id INT NOT NULL,  
    team_name VARCHAR(50),  
    PRIMARY KEY (team_id, person_id),  
    FOREIGN KEY (team_id) REFERENCES team(team_id) ON DELETE CASCADE,  
    FOREIGN KEY (person_id) REFERENCES person(person_id) ON DELETE CASCADE  
);
```

Front-End & Back-End source code: <https://cmu.to/Motoropedia-Project>

CRUD queries

From this page to the last covers the CRUD queries that are used in the project. The ‘?’ sign represents the placeholder of the insertion.

CREATE

Add Racer

- INSERT INTO person (first_name, last_name, status, date_of_birth, nationality, person_type) VALUES (?, ?, ?, ?, ?, ?);
- INSERT INTO racer (person_id, racer_license) VALUES (?, ?);

Add Tournament

- INSERT INTO person (tournament_name, date_of_match, circuit_street, circuit_city, circuit_state, circuit_zip, average_viewer_count, motorsport_id, caster_id, referee_id) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?);

Add Team

- INSERT INTO team (team_name, sponsor, country, win_count)VALUES (?, ?, ?, ?);

Add Staff Referee/Caster

- INSERT INTO person (first_name, last_name, status, date_of_birth, nationality, person_type) VALUES (?, ?, ?, ?, ?, ?);
- INSERT INTO staff (person_id, years_experience, staff_type, referee_license, language) VALUES (?, ?, ?, ?, ?);

Add Car

- INSERT INTO car (car_type, engine, manufacturer, product_year) VALUES (?, ?, ?, ?);

READ

Get Tournaments

- Get Tournaments

```
SELECT * FROM tournaments ORDER BY tournament_name ASC
```
- Get by tournament Id

```
SELECT t.tournament_id, t.tournament_name, t.date_of_match, t.circuit_street, t.circuit_city,
t.circuit_state, t.circuit_zip, t.average_viewer_count, t.caster_id, t.referee_id, t.motorsport_id,
m.motorsport_type, m.terrain

FROM tournaments t JOIN motorsport m ON t.motorsport_id = m.motorsport_id

WHERE t.tournament_id = ?
```
- Get by team Id

```
SELECT tm.tournament_id, tm.tournament_name, tm.date_of_match, tp.placement,
tp.average_laps_time, m.motorsport_type, tm.motorsport_id FROM tournaments tm INNER JOIN
tournament_participating tp ON tm.tournament_id = tp.tournament_id INNER JOIN team t ON tp.team_id
= t.team_id INNER JOIN motorsport m ON tm.motorsport_id = m.motorsport_id WHERE t.team_id = ?
ORDER BY date_of_match DESC
```
- Get by person Id

```
SELECT tm.tournament_id, tm.tournament_name, tm.date_of_match, tm.circuit_street, tm.circuit_city,
tm.circuit_state, tm.circuit_zip, tm.average_viewer_count, tm.motorsport_id, tm.caster_id,
tm.referee_id FROM tournaments tm
INNER JOIN tournament_participating tp ON tm.tournament_id = tp.tournament_id
INNER JOIN team_roster tr ON tp.team_id = tr.team_id WHERE tr.person_id = ? OR tm.caster_id = ? OR
tm.referee_id = ? ORDER BY tm.tournament_name ASC
```
- Get by motorsport Id

```
SELECT t.tournament_id, t.tournament_name, t.date_of_match, t.circuit_street, t.circuit_city,
t.circuit_state, t.circuit_zip, t.average_viewer_count, t.caster_id, t.referee_id, t.motorsport_id FROM
tournaments t
WHERE t.motorsport_id = ?
```

Get Car

- Get Car

```
SELECT * FROM car
```
- Get Car by motorsport Id

```
SELECT DISTINCT c.carmodel_id, c.car_type, c.engine, c.manufacturer, c.product_year, tp.team_id,
tm.team_name , m.motorsport_id FROM car c
INNER JOIN tournament_participating tp ON c.team_id = tp.team_id
INNER JOIN team tm ON tp.team_id = tm.team_id
INNER JOIN tournaments t ON t.tournament_id = tp.tournament_id
INNER JOIN motorsport m ON m.motorsport_id = t.motorsport_id
WHERE m.motorsport_id = ?
```
- Get Car by team Id

```
SELECT c.carmodel_id, c.car_type, c.engine, c.manufacturer, c.product_year, c.team_id FROM car c
WHERE c.team_id = ?
```
- Get Car by carmodel Id

```
SELECT c.carmodel_id, c.car_type, c.engine, c.manufacturer, c.product_year, t.team_name, t.team_id
FROM car c INNER JOIN team t ON c.team_id = t.team_id WHERE c.carmodel_id = ?
```

Get Team Roster

- Get Team Roster

```
SELECT * FROM team_roster
```
- Get Team Roster by team Id

```
SELECT t.team_id, p.person_id, p.first_name, p.last_name, p.status, p.date_of_birth, p.nationality,
p.person_type FROM team_roster tr
INNER JOIN team t ON tr.team_id = t.team_id
INNER JOIN person p ON tr.person_id = p.person_id
WHERE t.team_id = ?
```

Get Tournament

- Get Tournament
SELECT * FROM tournaments ORDER BY tournament_name ASC
- Get Tournament by tournament Id

SELECT t.tournament_id, t.tournament_name, t.date_of_match, t.circuit_street, t.circuit_city, t.circuit_state, t.circuit_zip, t.average_viewer_count, t.caster_id, t.referee_id, t.motorsport_id, m.motorsport_type, m.terrain FROM tournaments t JOIN motorsport m ON t.motorsport_id = m.motorsport_id WHERE t.tournament_id = ?
- Get Tournament by team Id
SELECT tm.tournament_id, tm.tournament_name, tm.date_of_match, tp.placement, tp.average_laps_time, m.motorsport_type, tm.motorsport_id FROM tournaments tm INNER JOIN tournament_participating tp ON tm.tournament_id = tp.tournament_id INNER JOIN team t ON tp.team_id = t.team_id INNER JOIN motorsport m ON tm.motorsport_id = m.motorsport_id WHERE t.team_id = ? ORDER BY date_of_match DESC
- Get Tournament by person Id
SELECT tm.tournament_id, tm.tournament_name, tm.date_of_match, tm.circuit_street, tm.circuit_city, tm.circuit_state, tm.circuit_zip, tm.average_viewer_count, tm.motorsport_id, tm.caster_id, tm.referee_id FROM tournaments tm INNER JOIN tournament_participating tp ON tm.tournament_id = tp.tournament_id INNER JOIN team_roster tr ON tp.team_id = tr.team_id WHERE tr.person_id = ? OR tm.caster_id = ? OR tm.referee_id = ? ORDER BY tm.tournament_name ASC
- Get Tournament by motorsport Id
SELECT t.tournament_id, t.tournament_name, t.date_of_match, t.circuit_street, t.circuit_city, t.circuit_state, t.circuit_zip, t.average_viewer_count, t.caster_id, t.referee_id, t.motorsport_id FROM tournaments t WHERE t.motorsport_id = ?

Get Motorsport

- Get Motorsport
SELECT * FROM motorsport
- Get Motorsport Type
SELECT * FROM motorsport WHERE motorsport_type = ?

Get Team

- Get Team
SELECT * FROM team ORDER BY team_name ASC
- Get Team by team Id
SELECT t.team_id, t.team_name, t.sponsor, t.country, t.win_count FROM team t WHERE t.team_id = ? ORDER BY t.team_name ASC
- Get Team by person Id
SELECT t.team_id, t.team_name, t.country, t.sponsor, t.win_count FROM team t INNER JOIN team_roster tr ON t.team_id = tr.team_id WHERE tr.person_id = ?
- Get Team by motorsport Id
SELECT DISTINCT t.team_id, t.team_name, t.sponsor, t.country, t.win_count, m.motorsport_id FROM team t INNER JOIN tournament_participating tp ON tp.team_id = t.team_id INNER JOIN tournaments tm ON tm.tournament_id = tp.tournament_id INNER JOIN motorsport m ON tm.motorsport_id = m.motorsport_id WHERE m.motorsport_id = ? ORDER BY t.team_name ASC

Get Tournament Participating

- Get Tournament participating
SELECT * FROM tournament_participating
- Get Tournament participating by tournament Id
SELECT tp.tournament_id, tp.team_id, t.team_name, tp.average_laps_time, tp.start_time, tp.finish_time, tp.placement FROM tournament_participating tp INNER JOIN team t ON tp.team_id = t.team_id WHERE tp.tournament_id = 20001 ORDER BY tp.placement ASC

Get Person

- Get Person by person Id
SELECT p.person_id, p.first_name, p.last_name, p.status, p.date_of_birth, p.nationality, person_type
FROM person p
WHERE p.person_id = ?

Get Racer

- Get all Racer
SELECT p.person_id, p.first_name, p.last_name, p.status, p.date_of_birth, p.nationality, r.racer_license
FROM racer r
INNER JOIN person p ON r.person_id = p.person_id
- Get Racer by motorsport Id
SELECT DISTINCT p.person_id, p.first_name, p.last_name, p.status, p.date_of_birth, p.nationality,
p.person_type, r.racer_license FROM person p
INNER JOIN team_roster tr ON p.person_id = tr.person_id
INNER JOIN tournament_participating tp ON tp.team_id = tr.team_id
INNER JOIN tournaments tm ON tp.tournament_id = tm.tournament_id
INNER JOIN motorsport m ON m.motorsport_id = tm.motorsport_id
INNER JOIN racer r ON p.person_id = r.person_id WHERE m.motorsport_id = ?
ORDER BY p.first_name ASC

Get Staff

- Get all Staff
SELECT p.first_name, p.last_name, p.status, p.nationality, p.date_of_birth, s.years_experience,
s.staff_type, s.referee_license, s.language FROM person p
INNER JOIN staff s ON p.person_id = s.person_id
ORDER BY p.first_name
- Get Staff by person Id
SELECT * FROM person p INNER JOIN staff s ON p.person_id = s.person_id
WHERE p.person_id = ?
- Get all Referee
SELECT p.person_id, p.first_name, p.last_name, p.status, p.nationality, s.years_experience,
s.referee_license FROM person p
INNER JOIN staff s ON p.person_id = s.person_id
WHERE s.staff_type = 'Referee'
ORDER BY p.first_name ASC
- Get all Caster
SELECT p.person_id, p.first_name, p.last_name, p.status, p.nationality, s.years_experience, s.language
FROM person p
INNER JOIN staff s ON p.person_id = s.person_id
WHERE s.staff_type = 'Caster'
ORDER BY p.first_name ASC

Get Car, Racer, Team,Tournament, and Staff by **Keyword**

- Get Car
SELECT DISTINCT c.carmodel_id, c.car_type, c.engine, c.manufacturer, c.product_year, t.team_id, t.team_name FROM car c
INNER JOIN team t ON c.team_id = t.team_id
WHERE (CONCAT(c.car_type, ' ', c.engine, ' ', c.manufacturer, ' ', c.product_year)) LIKE ?
ORDER BY c.car_type ASC
- Get Racer
SELECT DISTINCT p.person_id, p.first_name, p.last_name, p.status, r.racer_license, p.date_of_birth, p.nationality FROM person p
INNER JOIN racer r ON p.person_id = r.person_id
WHERE (CONCAT(p.first_name, ' ', p.last_name) LIKE ?)
ORDER BY p.first_name ASC;
- Get Team
SELECT DISTINCT t.team_id, t.team_name, t.sponsor, t.country, t.win_count FROM team t
WHERE t.team_name LIKE ?
ORDER BY t.team_name ASC
- Get Tournament
SELECT DISTINCT t.tournament_id, t.tournament_name, t.date_of_match, t.average_viewer_count, t.circuit_street, t.circuit_city, t.circuit_state, t.circuit_zip, m.motorsport_type, m.motorsport_id, t.referee_id, t.caster_id FROM tournaments t
INNER JOIN motorsport m ON m.motorsport_id = t.motorsport_id
WHERE tournament_name LIKE ? OR circuit_state LIKE ?
ORDER BY t.tournament_name ASC
- Get Staff
SELECT p.person_id, p.first_name, p.last_name, p.date_of_birth, p.nationality, s.staff_type, s.years_experience, p.status, s.referee_license, s.language FROM person p
INNER JOIN staff s ON p.person_id = s.person_id
WHERE (CONCAT(p.first_name, ' ', p.last_name)) LIKE ?
ORDER BY p.first_name ASC

UPDATE

Update Racer

UPDATE person SET first_name = ?, last_name = ?, status = ?, date_of_birth = ?, nationality = ?
WHERE person_id = ?

Update Tournament

UPDATE tournaments SET tournament_name = ?, date_of_match = ?, circuit_street = ?, circuit_city = ?, circuit_state = ?, circuit_zip = ?, average_viewer_count = ?, motorsport_id = ?, caster_id = ?, referee_id = ?
WHERE tournament_id = ?

Update Team

UPDATE team SET team_name = ?, sponsor = ?, country = ?, win_count = ?
WHERE team_id = ?

Update Staff

UPDATE staff SET years_experience = ?, staff_type = ?, referee_license = ?, language = ?
WHERE person_id = ?

DELETE

Delete Racer by person Id

DELETE FROM person p WHERE p.person_id = ?

Delete Tournament by tournament Id

DELETE FROM tournaments t WHERE t.tournament_id = ?

Delete Team by team Id

DELETE FROM team t WHERE t.team_id = ?

Delete Staff by person Id

DELETE FROM staff s WHERE s.person_id = ?

Delete Car by carmodel Id

DELETE FROM car c WHERE c.carmodel_id = ?

Data gathering method:

We use AI(Gemini) to generate our data, some data is real-world data, but some is generated data.

Prompt:

"Generate a complete SQL script with INSERT statements to populate the database defined by the motoropedia.sql schema I've provided. The script must generate a large and comprehensive dataset for the year [Specify Year Here].

Crucially, you must first analyze the CREATE TABLE statements in the provided schema to determine the exact number of columns for each table. Every INSERT statement you write must provide a value for each and every column and must not contain more values than the columns defined for that table.

The data should cover the following 12 motorsport types: Autocross, Autotest, Circuit Racing, Cross Country, Drag Racing, Drifting, Hill Climb, Karting, Rallying, Rally Cross, Sprint, and Trials.

Follow these specific instructions for each table to ensure the data is realistic, consistent, and adheres strictly to the schema:

General Rules

No Null Values: Every column in every table must have a value.

Data Volume: Generate at least 50 teams, 200 persons (racers and staff), 25 tournaments, and 100 cars. Create a rich set of participation records to connect them.

ID Integrity: All primary and foreign key IDs must be unique, sequential, and correctly referenced across tables.

Table-Specific Instructions

motorsport (3 columns)

Create one entry for each of the 12 specified motorsport types (motorsport_type).

Assign a plausible terrain for each type (e.g., 'Asphalt/Tarmac' for Circuit Racing).

team (5 columns)

Generate realistic team_name and sponsor combinations using real-world examples (e.g., Team Penske/Shell, Hyundai Motorsport/Shell Mobis).

Distribute the teams across various countries (country).

Generate a plausible win_count.

person (7 columns)

Racers: Use the real names (first_name, last_name) of at least 100 famous racers (e.g., Lewis Hamilton, Sébastien Loeb, Ken Block). Set their person_type to 'racer'.

Staff: Generate realistic but fictional names for at least 100 staff members. Set their person_type to 'staff'.

Populate status, date_of_birth, and nationality for each person.

racer (2 columns)

For every person with person_type = 'racer', create a corresponding entry with their person_id and a unique racer_license.

staff (5 columns)

For every person with person_type = 'staff', create a corresponding entry.

The staff_type must be either 'Referee' or 'Caster'.

The referee_license must be a unique number for referees and can be '0' for casters.

Populate years_experience and language.

team_roster (3 columns)

Assign each person (person_id) to a team (team_id).

Ensure the team_name matches the name in the team table for the corresponding team_id.

car (6 columns)

For each team, create at least two car models.

The car_type, engine, and manufacturer should be appropriate for the team's primary motorsport discipline.

The product_year must match the year specified for the dataset.

tournaments (8 columns)

Create a variety of tournaments for each of the 12 motorsport types.

Use realistic or plausible tournament names (e.g., 'Daytona 500', 'Rally Finland') for the circuit_street column.

Populate the location details in circuit_city and circuit_state.

The date_of_match must fall within the year specified for the dataset.

tournament_participating (8 columns)

Create numerous records to link tournaments and teams.

For each record, assign a staff_id of a person whose staff_type is 'Referee'.

Generate plausible performance data: laps_time, placement, and viewer_count.

Set start_date and finish_date that align with the date_of_match in the tournaments table.

Please generate the final output as a single, executable .sql file."